Sharma and Sarode *Discover Artificial Intelligence* (2026) 6:246
https://doi.org/10.1007/s44163-026-01013-9

PERSPECTIVE Open Access

Human-assisted traditional peer review intelligence vs artificial peer review intelligence: perspectives on future disruptive sciences

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Abstract
Human-assisted Traditional Peer Review Intelligence (HATPRINT) plays a crucial role in advancing scientific discoveries through critical evaluation and expert insights by humans. However, the rise of Artificial Intelligence-assisted Peer Review Intelligence (AIPRINT), powered by Generative AI (Gen AI), including various forms of LLMs is likely to transform the peer review landscape. While Gen AI can enhance efficiency, automate reviews, and refine research, overreliance may risk losing innovation and perpetuate pseudo-peer review cycles. Several AI-assisted review systems, such as SciScore, Paperpal, and various other publishers in-house AI reviewer suggestion tools, suggest futurist dominance of AIPRINT over HATPRINT. These platforms could enhance, rather than replace, human evaluation by automating checks, improving writing, and aiding reviewer selection. Theoretically, their integration could lead to the growing convergence of HATPRINT and AIPRINT leading to a semi-autonomous peer review systems strongly underscoring the need for governance, transparency, and human oversight. We propose a balanced convergence of HATPRINT and AIPRINT to maintain scientific integrity, ensuring human expertise remains integral. However, explicit limitations are lack for empirical analysis in support of the proposed viewpoints. Future research should explore this equilibrium to sustain disruptive science while leveraging AI for enhanced accuracy, productivity, and the evolution of peer review.

Keywords Human-assisted traditional peer review intelligence (HATPRINT), Artificial peer review intelligence (AIPRINT), Disruptive sciences, Journals, Publications, Research, Sustainability

1 Introduction

In broader perspectives, human intelligence is attributed to the general mental ability for reasoning, problem-solving, learning, self-awareness, and integration of cognitive functions such as perception, emotions, attention, memory, language, planning, comprehension, critical thinking, creativity, abstraction, logic, and understanding of facts and data [1–3].

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Among the various forms of human intelligence, the scientific peer review process embodies key attributes such as comprehension, critical thinking, creativity, abstraction, logic, and the interpretation of facts and data [2–5]. Human-assisted traditional peer review intelligence (HATPRINT) represents human-centered evaluative intelligence characterized by expert judgment, contextual understanding, analytical reasoning, creativity, and ethical discernment applied during assessment of research manuscripts and innovations.

In the past, HATPRINT has ensured rigor, accuracy, and credibility in scientific publishing, shaping the evolution of disruptive science. Beyond publication review, it also underpins the evaluation of research projects, their outcomes (such as papers and patents), and the identification of gaps that can guide future investigations toward more innovative discoveries.

It is important to discuss how Generative Artificial Intelligence (Gen AI), including Large Language Models (LLMs) such as OpenAI's GPT series (ChatGPT), Google's Gemini, Anthropic's Claude, and Meta's Llama, alongside powerful open-source options like Mixtral, Mistral, DeepSeek, and Falcon etc. will shape HATPRINT and its role in driving disruptive scientific discoveries. Understanding both the benefits and challenges of Gen AI is crucial for achieving better outcomes by HATPRINT in advancing disruptive science and building a sustainable society. The use of Gen AI tools might be helpful for the peers/reviewers to channel their thought processes to come up with productive comments [5–10]. Besides these notable uses, Gen AI and LLMs are reported multiple functionalities such as fake-new judgment and fact-check [4, 5].

It is important to examine how the evolution of Gen AI and LLMs may play a crucial role on behalf of Peers/reviewers (Humans), leading to a form of Artificial Peer Review Intelligence (AIPRINT) [10–15]. Gen AI and LLMs will probably evolve to play the role of Artificial Peer Reviewer. Then, it would be rational to explore the implications of AIPRINT in the future path of disruptive sciences and innovations [15–20]. AIPRINT can be denoted as Gen AI and LLMs powered peer review intelligence to review scientific documents such as manuscripts, articles, research projects, research progress reports, research outputs, and innovations by comprehension, abstraction, logic, thinking, and understanding of facts and data.

However, with the increasing potential of AIPRINT, potential risks of bias, superficial analysis, and ethical challenges could emerge in the long run if reliance is exclusively on AIPRINT. A probability of emerging a form of peer review intelligence as Pseudo Peer Review Intelligence (PSEUDOPRINT) due to the continuous vicious cycle of HATPRINT-APRINT.

In the future, the convergence of HATPRINT and AIPRINT needs to be studied and evaluated carefully, as it could lead to a potential issue, such as the escape velocity phenomenon of peer review processes. Convergence of human machine is proposed to refer to the integration of HATPRINT with AIPRINT to create a balanced and collaborative model. In this state, LLMs manage routine and large-scale analysis, while humans contribute creativity, ethical judgment, and contextual expertise, together strengthening the efficiency and integrity of peer review. In the last stage, the escape velocity phenomenon is suggested to describe the tipping point at which AIPRINT surpasses HATPRINT, leading to primarily AI-driven peer review processes. At this stage, peer review risks becoming machine-dominated, raising questions about the role of human expertise,

creativity, and accountability in sustaining disruptive science. This progressive interaction is schematically illustrated in Fig. 1.

This perspective is presented as a theoretical exploration (not an experimental or systematic review) on HATPRINT vs. AIPRINT that highlights the crucial role of humans as peers and reviewers in convergence with Gen AI to advance disruptive knowledge, inventions, and discoveries through critical analysis, suggestions, and insights. This paper evaluates the relevance, implications, risks, and ethical concerns of AIPRINT in the context of emerging technologies like Gen AI for a sustainable and converged scientific future.

2 Relevance of disruptive peer review intelligence (HATPRINT)

HATPRINT functions as a feedback mechanism that refines methodologies, strengthens arguments, and supports the emergence of disruptive discoveries. HATPRINT contributes through contextual critique and ethical oversight. By offering diverse perspectives and mentorship, experienced reviewers help advance scientific progress and guide the next generation of researchers [15–20]. Besides above role in the disruptive discoveries and sciences, the role of HATPRINT in ethical, social, cultural and diversity-based judgement beyond technical and process driven validation is crucial.

2.1 Artificial peer review intelligence and its impact

AIPRINT can be referred to as an emerging paradigm of peer review intelligence, mainly driven by Gen AI-based LLMs. Unlike human-led HATPRINT, AIPRINT operates by processing vast datasets, applying algorithmic logic, and automating review processes

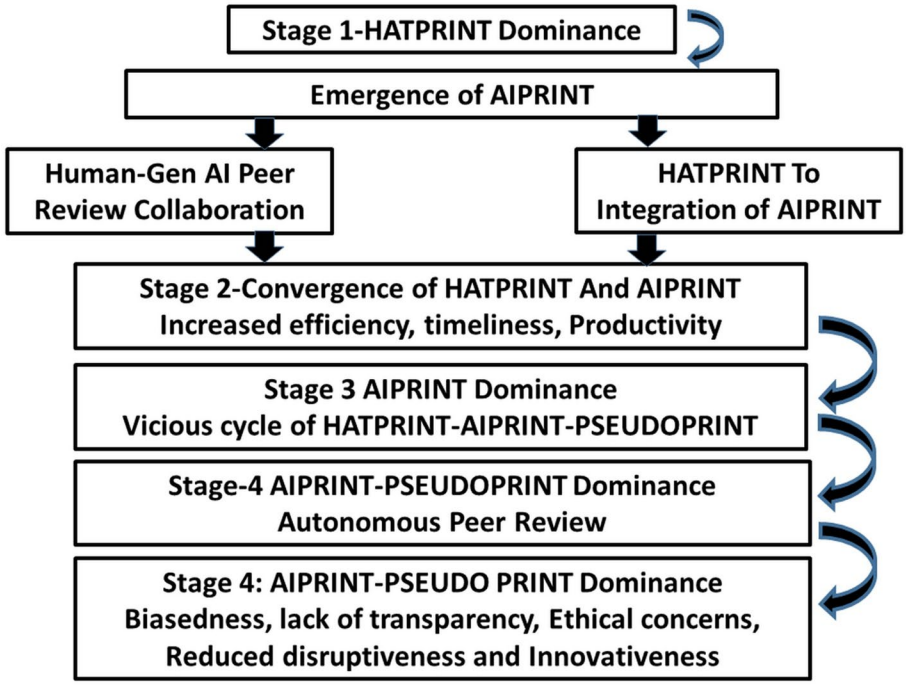

Fig. 1 Stages of transition from Human-assisted traditional peer review intelligence (HATPRINT) to Artificial Intelligence-Assisted Peer Review Intelligence (AIPRINT) and Pseudo Peer Review Intelligence (PSEUDOPRINT) in peer review. Each stage represents a conceptual shift in the locus of evaluative authority—from predominantly human judgment (Stage 1) to increasing AI assistance (Stage 2), followed by potential AI dominance (Stage 3) and risks of PSEUDOPRINT (Stage 4). The arrows denote bidirectional interaction rather than linear replacement

with speed, scalability, and efficiency. AIPRINT has the potential to streamline editorial workflows, reduce reviewer fatigue, and rapidly detect errors, inconsistencies, or plagiarism. However, it also carries risks of superficial evaluation, loss of critical depth, algorithmic bias, and ethical concerns such as a lack of transparency and accountability. In its ideal form, AIPRINT functions as a complementary driver of peer review, augmenting human expertise while accelerating efficiency. In its advanced state, it may evolve toward autonomous peer review systems, potentially surpassing HATPRINT in influence a shift described in the manuscript as the "escape velocity phenomenon."

An ideal symbiotic and successful collaboration between HATPRINT and AIPRINT is expected as AIPRINT is the role of the driver of peer review processes and the primary driver of the peer review process and HATPRINT's role in context and expertise. The increasing convergence of HATPRINT and AIPRINT will likely impact authors, researchers, and scholars in various ways, with adaption to AI-driven peer review processes, augmented Human-AI collaboration will become more prevalent, enabling researchers to focus on complex tasks [15–20].

In the future development of advanced AIPRINT tools that can enhance the accuracy and efficiency of AI-powered peer review tools by addressing concerns surrounding bias, transparency, and accountability in AI-driven peer review processes [21–25]. Gen AI can enhance HATPRINT by automating routine tasks, providing preliminary reviews, and identifying potential areas for improvement. While AIPRINT-driven tools may change the nature of peer review, with AI-assisted reviews becoming more prevalent. A summarized comparison between HATPRINT and AIPRINT is represented in Table 1.

3 Pseudo peer review intelligence

Pseudo Peer Review Intelligence (PSEUDOPRINT) can be described as a pseudo-form of peer review in which AI-generated insights may potentially dominate but are presented as human evaluations. This hybrid model blurs the boundary between HATPRINT and AI-derived assessments, thereby undermining transparency, credibility, and accountability in scholarly and scientific evaluations. In this form, the peer review process is formally attributed to human experts, yet the substantive intellectual contribution is largely influenced or even driven by Gen AI tools and large language models. Although PSEUDOPRINT may appear to preserve the features of HATPRINT, it effectively shifts the evaluative process toward AIPRINT. This shift in the convergence of HATPRINT and AIRPRINT could influence accountability, impacting innovation, and introducing ethical challenges [18–25].

4 The convergence cycle between HATPRINT, AIPRINT and PSEUDOPRINT

The convergence of HATPRINT and AIPRINT refers to the progressive integration of human expertise with artificial intelligence–driven peer review processes. Rather than existing in isolation, both systems increasingly interact to complement one another: HATPRINT contributes creativity, contextual knowledge, ethical judgment, and nuanced critique, while AIPRINT adds speed, scalability, automation, and the ability to process vast amounts of literature. This convergence is envisioned as a symbiotic collaboration in which humans and machines collectively shape a more efficient, accurate, and sustainable peer review ecosystem. In this model, AIPRINT handles routine and

Table 1 Comparative summary of Human-assisted traditional peer review intelligence (HATPRINT) and artificial intelligence peer review intelligence (AIPRINT)

Aspect	Human-assisted traditional peer review intelligence (HATPRINT)	Artificial intelligence peer review intelligence (AIPRINT)
Key characteristics	Based on human expertise, critical thinking, abstraction, creativity, and comprehension. Provides contextual, field-specific insights. Integrates cognitive and emotional intelligence. Relies on lived academic and research experience.	Driven by Generative AI (LLMs, Chatbots) and future AGI. Automates review processes and accelerates feedback cycles. Can process vast literature instantly. Relies on algorithms, data patterns, and training corpora.
Strengths	Ensures rigor, accuracy, and credibility of scientific review. Offers diverse perspectives and creative interpretations. Provides mentorship and guidance to junior researchers. Capable of detecting nuanced flaws, ethical issues, and contextual gaps. Quantitative or procedural metrics (e.g. error detection rates are more based on the reality, nuanced judgment and consciences)	Enhances efficiency and scalability of peer review. Reduces reviewer fatigue and time delays. Identifies technical errors, inconsistencies, and plagiarism rapidly. Potential for bias detection and objectivity in evaluation. Quantitative or procedural metrics (e.g. turnaround times is reduced)
Risks/Limitations	Subject to reviewer fatigue, bias, or conflicts of interest. Limited scalability due to human availability. Variability in review quality and depth. Slower turnaround times compared to AI-driven reviews. Quantitative or procedural metrics (e.g. turnaround times is lengthened due to human linked behaviors)	Risk of superficial or mechanical reviews lacking deep critical analysis. May overlook disruptive or unconventional ideas. Vulnerable to bias inherited from training data. Ethical concerns: transparency, accountability, and pseudo-peer review cycles. Over-reliance could marginalize human expertise, leading to innovation loss. Quantitative or procedural metrics (AIPRINT showing high accuracy for technical checks but significant limitations for nuanced academic judgments)

repetitive tasks, freeing human reviewers to focus on higher-order critical thinking and disruptive insights. Convergence is thus not a simple replacement of human intelligence by AI, but a balanced partnership where the strengths of both are leveraged for disruptive and sustainable science.

Disruptive researchers, authors, innovators, and scholars commonly receive suggestions, insights, creative viewpoints, interpretations, analysis, and other forms of human intelligence attributes that contribute to shaping the present and future path of disruptive sciences [10–15]. However, in the emerging scenarios of uses of Artificial Peer Intelligence assistance to disruptive peer intelligence, suggestions, insights, creative viewpoints, interpretations, analysis, and other forms of inputs will be included in the revised draft of disruptive researchers, authors, innovators, and scholars [15–20]. The convergence of human-assisted disruptive peer review intelligence (HATPRINT) and artificial peer review intelligence (AIPRINT) is a plausible scenario. This convergence could lead to enhanced, productive, and processed forms of peer review intelligence (Fig. 1).

Cyclic interaction among HATPRINT, AIPRINT, and PSEUDOPRINT may generate increasingly autonomous peer review systems. This transition is conceptualized as the escape velocity phenomenon. This evolving cycle could generate novel versions of AIPRINT applicable across basic, translational, and clinical sciences, contributing to

a technologically sustainable research ecosystem. The authors predict that with ongoing convergence, the interference of AIPRINT in traditionally human-assisted domains will intensify, potentially reaching a stage where AIPRINT surpasses HATPRINT in both influence and functionality. At this advanced phase, human contributions such as abstraction, comprehension, analytical judgment, and creativity may become secondary, as LLMs assume primary control of evaluative processes. Consequently, the peer review pathway may evolve toward an AI-dominated paradigm, underscoring the need for proactive governance to ensure transparency, accountability, and sustained human oversight in future peer review ecosystems.

A vicious cycle of HATPRINT-AIPRINT could result in increased efficiency, automation, and human assistance to streamline the peer review process, improve accuracy, enhance the detection of errors and inconsistencies, and enhance productivity. The escape velocity phenomenon in the context of the convergence of HATPRINT and AIPRINT can refer to the peer review intelligence point at which AIPRINT surpasses HATPRINT to achieve revolutionized forms of peer review processes. At this point, the HATPRINT role by humans has to be adaptive, collaborative, and symbiotic for a sustainable future world.

The "escape velocity" phenomenon in peer review intelligence refers to the critical threshold at which AIPRINT evolving into forms such as PSEUDOPRINT, surpass HATPRINT in influence and dominance. At this stage, the peer review process begins to operate autonomously, with minimal human intervention, creating a self-sustaining cycle of AI-based evaluation. In the cyclic continuum of peer review, "escape velocity" thus represents the transition point where the system moves from human-assisted models (HATPRINT–AIPRINT cycles) toward predominantly AI-governed review mechanisms. The cyclic dynamics among HATPRINT, AIPRINT, and PSEUDOPRINT are depicted in Fig. 2, which conceptualizes the escape velocity phenomenon. At this stage, AI systems are capable of independently reviewing, refining, and even influencing scientific content, potentially reducing human roles to oversight or validation. While this phenomenon may lead to rapid scalability and efficiency, it also raises concerns about over-reliance on AI, loss of human creativity and contextual depth, and ethical accountability. In essence, the escape velocity phenomenon marks the tipping point where the peer review ecosystem risks shifting from a collaborative model to an AI-dominated one.

Several AI-assisted review tools and prototypes are already in place that support the early evolution of AIPRINT within current scholarly ecosystems. For instance, SciScore automates screening with regard to the reproducibility, transparency, and methodological aspects of research papers [26]. While Paperpal assists reviewers and authors for assistance in language help for scientific writing, readability, and logical flow through AI-powered contextual feedback [27]. Various AI-assisted plagiarism tools, such as Turnitin [28], Qillbot [29], Grammarly [30] assist in plagiarism check and provide suitable feedback to enhance the quality of texts. Furthermore, AI-assisted tools for creating and enhancing the publication images, such as Illustrae [31] and BioRender [32] are helpful for the researchers to compile and present their data. Similarly, Elsevier's reviewer suggestion system employs machine learning to match manuscripts with suitable reviewers, and Springer Nature, IEEE, and Frontiers are experimenting with AI screening tools to identify statistical inconsistencies and plagiarism. These examples represent the initial stages of AIPRINT, where automation enhances, but does not replace, human evaluative

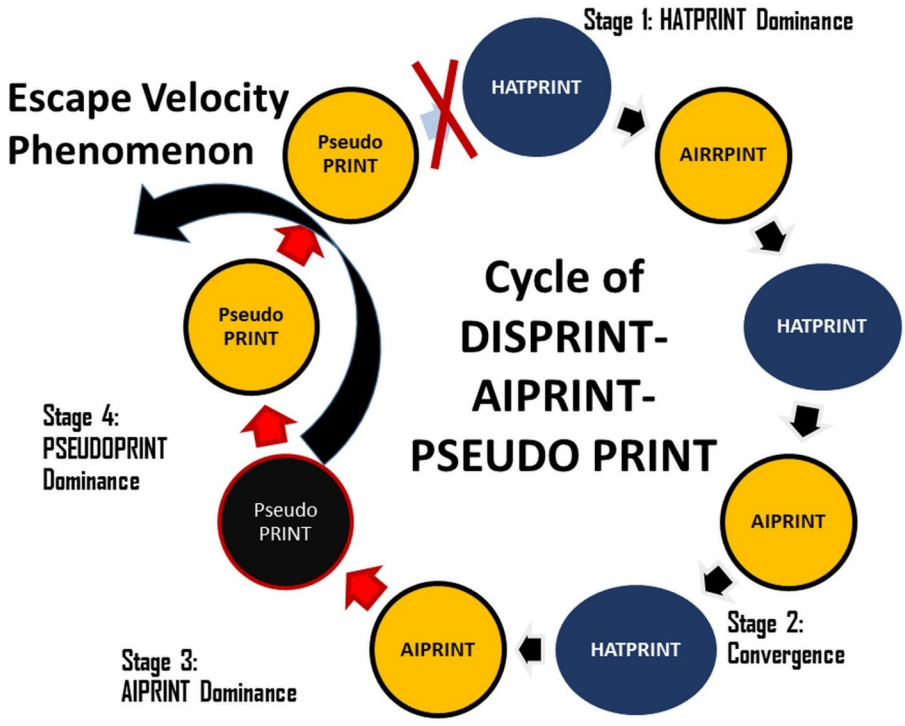

Fig. 2 The cyclic progression and escape velocity phenomenon in Human-assisted traditional peer review intelligence (HATPRINT), Artificial Intelligence-Assisted Peer Review Intelligence (AIPRINT), and Pseudo Peer Review Intelligence (PSEUDOPRINT) systems. The circular arrows illustrate iterative feedback among HATPRINT, AIPRINT, and PSEUDOPRINT, while the outward arrow represents the escape velocity threshold beyond which AI-driven review may become self-sustaining with limited human oversight

intelligence. Integrating these tools within editorial workflows demonstrates a growing convergence of HATPRINT and AIPRINT. The proposed manuscript aligns with this transition by conceptualizing how such AI-driven systems could evolve from assistive roles toward autonomous or semi-autonomous review processes, potentially culminating in the "escape velocity phenomenon" described herein. This reflection situates the manuscript's theoretical framework within ongoing real-world developments, emphasizing the need for balanced governance, transparency, and sustained human oversight to ensure integrity and disruptiveness in future scientific reviews. In summary, this information is in support of argument rather than replacing human review.

5 Challenges and ethical concerns with artificial peer review intelligence (AIPRINT)

Risks associated with Stage 3–4 transitions shown in Fig. 1 underpin the ethical challenges discussed below. As the convergence of HATPRINT and AIPRINT accelerates, future ethical considerations should look beyond immediate concerns of transparency, bias, and errors to encompass comprehensive issues of accountability, authorship, and trust in disruptive sciences. The emergence of PSEUDOPRINT could lead to a lack of authenticity of scholarly evaluation if AI contributions are misrepresented as human judgment, and also pinpoint ethical concerns on levels and involved stakeholders. With regard to AIPRINT, there should be procedures to safeguard against systemic biases that may disproportionately privilege certain disciplines or geographies. Furthermore, ethical stewardship will require a dynamic equilibrium where human creativity, contextual

insight, and moral responsibility remain central, while AI contributes scalability and efficiency. Anticipating the "escape velocity phenomenon," in which AIPRINT could surpass HATPRINT dominance, ethical foresight demands proactive policies that ensure peer review remains a collaborative, accountable, and trustworthy enterprise in shaping future sustainable and disruptive sciences. In medical sciences, possible forms of ethical concerns may emerge if AIRPRINT and PSEUDOPRINT-based peer review insights and suggestions could be contributory components for the failures of drugs and therapies against diseases.

AIPRINT could become a reality, with Gen AI and LLMs capable of simulating human-like peer review processes [25, 33–36]. Over-reliance may lead to overlooking innovative, unconventional ideas. Gen AI and LLMs may struggle to fully comprehend complex research contexts. AIPRINT may lead to superficial reviews and a lack of critical analysis and constructive criticism, biased and unfair evaluation of scientific documents. If current trends accelerate unchecked, we may envision a future where disruptive scientific pathways are predominantly AI-driven, potentially marginalizing traditional human-led innovation.

6 AI governance in peer review

Responsible AI governance in peer review requires balancing efficiency and timeliness with the preservation of scientific integrity and innovation. The rise of AIPRINT and its convergence with HATPRINT offer benefits such as scalability, improved error detection, and reduced workload for reviewers. However, risks including algorithmic bias, superficial assessment, and PSEUDOPRINT highlight the need for strong governance frameworks.

Unchecked use of AI in scholarly review may amplify systemic bias, reduce accountability, and obscure transparency, especially if AI-generated insights are misrepresented as human judgments. Governance structures must therefore ensure transparency through mandatory disclosure of AI involvement, human oversight, and regular auditing of algorithmic decisions. To prevent the "escape velocity phenomenon," where AI surpasses human review in influence and erodes contextual depth, governance systems should define limits and monitor transitions. Institutions, research bodies, and policy agencies must work toward a symbiotic model integrating HATPRINT and AIPRINT, guided by ethical safeguards, accountability, and transparency to foster credible and disruptive scientific innovation.

A multi-tier governance framework is essential to guide the responsible use of AI and peer review intelligence across preclinical and clinical research domains. At the institutional level, Research Ethics Committees and Institutional Review Boards should integrate AI-audit and disclosure modules to evaluate the extent of AI assistance in data analysis, interpretation, and peer review. Governmental agencies, such as the Department of Science and Technology, the Indian Council of Medical Research, and the U.S. National Institutes of Health can establish policy directives mandating transparency, algorithmic explainability, and human oversight for AI-assisted reviews of preclinical and clinical studies. Internationally, organizations like the WHO, OECD, UNESCO, and the International Committee of Medical Journal Editors are positioned to harmonize global standards for AI governance, ensuring ethical consistency and interoperability of AI-based evaluation systems. International Research Societies and Committees

(e.g., COPE, WAME, and ISC) can play a pivotal role by framing accreditation systems and ethical codes for AI-involved peer review processes. Such a layered model spanning institutional, national, and global levels can safeguard scientific credibility, ensure compliance with research ethics, and prevent algorithmic dominance over human judgment in preclinical and clinical studies.

7 Conclusion

In conclusion, HATPRINT remains indispensable for sustaining scientific advancement. While Gen AI will increasingly influence peer review, it is unlikely to fully replace human expertise. Instead, the integration of AI-driven analysis can complement human evaluation by enhancing objectivity, detecting biases, and improving the efficiency of complex, data-intensive reviews.

As disruptive sciences continue to evolve, ethical principles such as transparency, conflict of interest management, and bias mitigation will become central to maintaining integrity within the converging landscape of HATPRINT and AIPRINT. The future of peer review is expected to adopt a collaborative human–AI model, that can leverage analytical precision alongside human contextual judgment and creativity.

Long-term studies will be needed to assess the equilibrium between HATPRINT and AIPRINT and to quantify their relative contributions within published research. Clear disclosure of AI involvement in peer review will guide researchers, innovators, and policymakers in interpreting the authenticity and credibility of scientific outputs.

In future, robust AI governance frameworks will be essential to create a sustainable and ethically sound peer review ecosystem one that balances productivity, transparency, and innovation while preventing the dominance of algorithmic evaluation.

The proposed terms HATPRINT, AIPRINT, and PSEUDOPRINT are original conceptualizations introduced to better articulate this evolving and convergent trajectory of peer review intelligence in scholarly and scientific communication.

Author contributions
"N.K.S. and S.C.S. wrote the main manuscript text and figures 1-2. All authors reviewed the manuscript."

Funding
There is no funding to declare.

Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and informed consent
Not applicable.

Consent to participate
Not applicable.

Declaration of generative AI and AI-assisted technologies in the writing process
During the preparation of this work, the authors used ChatGPT 5.0 and Grammarly in order to improve the language, better coherence and conciseness. After using these tools, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Competing interests
The authors declare that they have no conflict of interest.

Received: 4 September 2025 / Accepted: 9 February 2026
Published online: 18 February 2026

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